

# WEEK 10

## Title

**Mobile Number Validation using User-Defined Exception Handling**

## Objective

To develop a Java program to validate a mobile number using **try-catch**, **built-in exceptions** and a **user-defined exception**.

## System Requirements

### Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and Monitor

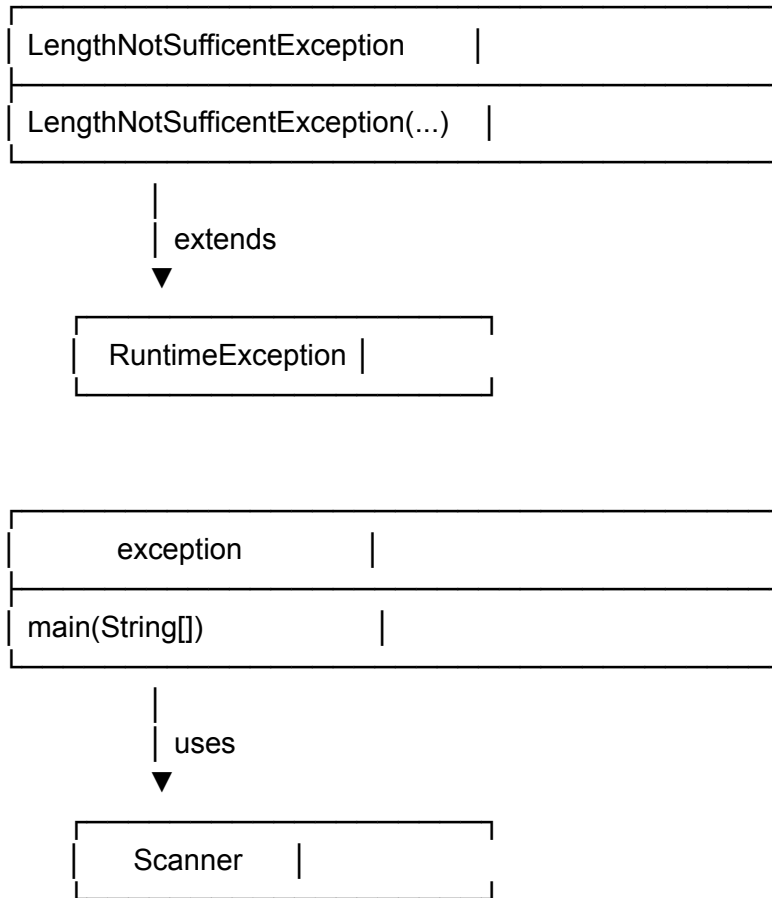
### Software Requirements

- JDK 8 or above
- VS Code / Eclipse / IntelliJ IDEA
- Command Prompt / PowerShell

## Algorithm / Logic Description

1. Create a user-defined exception named `LengthNotSufficientException`.
2. Extend the exception from `RuntimeException`.
3. Read the mobile number from the user using `Scanner`.
4. Check whether the length is greater than 10.
5. If the length is greater than 10, throw `ArrayIndexOutOfBoundsException`.
6. Check whether the length is less than 10.
7. If the length is less than 10, throw `LengthNotSufficientException`.
8. Convert the mobile number into a `long` using `Long.parseLong()`.
9. If the conversion is successful, display that the mobile number is valid.
10. Handle the exceptions using separate `catch` blocks.
11. Display suitable error messages.

## UML Class Diagram



## Source Code

Save the file as **exception.java**

```
import java.util.Scanner;
```

```
// User-defined exception
```

```
class LengthNotSufficientException extends RuntimeException {
```

```
    // Constructor
```

```
    LengthNotSufficientException(String message) {
        super(message);
    }
```

```
}
```

```
// Main Class
```

```
public class exception {
```

```

public static void main(String[] args) {

    // Creating Scanner object
    Scanner sc = new Scanner(System.in);

    System.out.println("Enter mobile number");
    String number = sc.nextLine();

    try {

        // Checking length greater than 10
        if (number.length() > 10) {
            throw new ArrayIndexOutOfBoundsException(
                "Mobile number should not exceed 10 digits"
            );
        }

        // Checking length less than 10
        if (number.length() < 10) {
            throw new LengthNotSufficientException(
                "Mobile number should contain 10 digits"
            );
        }

        // Converting String to long
        long n = Long.parseLong(number);

        System.out.println("Mobile number is valid");
    }

    // Handling user-defined exception
    catch (LengthNotSufficientException e) {
        System.out.println(e.getMessage());
    }

    // Handling built-in exception
    catch (ArrayIndexOutOfBoundsException e) {
        System.out.println(e.getMessage());
    }

    // Handling invalid number format
    catch (NumberFormatException e) {
        System.out.println("Enter only digits");
    }
}

```

```
        sc.close();  
    }  
}
```

## Compilation and Execution Commands

### Compile

```
javac exception.java
```

### Run

```
java exception
```

## Output

### Case 1: Valid Mobile Number

Enter mobile number

9876543210

Mobile number is valid

### Case 2: Less than 10 Digits

Enter mobile number

987654321

Mobile number should contain 10 digits

### Case 3: More than 10 Digits

Enter mobile number

98765432101

Mobile number should not exceed 10 digits

### Case 4: Characters Entered

Enter mobile number

98765abc10

Enter only digits

```
PS C:\Users\navee> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:57522' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\navee\AppData\Local\Temp\vscodesws_58665\jdt_ws\jdt.ls-java-project\bin' 'exception'
Enter mobile number
950WE345
Mobile number should contain 10 digits
PS C:\Users\navee> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:56260' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\navee\AppData\Local\Temp\vscodesws_58665\jdt_ws\jdt.ls-java-project\bin' 'exception'
Enter mobile number
QASWERFGTY
Enter only digits
PS C:\Users\navee> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:64209' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\navee\AppData\Local\Temp\vscodesws_58665\jdt_ws\jdt.ls-java-project\bin' 'exception'
Enter mobile number
9502621641
Mobile number is valid
PS C:\Users\navee> █
```